# Lakshya Jindal

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## Education

**International Institute of Information Technology, Hyderabad** Aug 2023 – Present
*B.Tech + M.S. (By Research) in Electronics and Communication Engineering* *Hyderabad, India*

**Hiranandani Foundation School Thane** June 2021 – June 2023
*ISC, PCM and Computer Applications - 94.25%* *Mumbai, India*

## Experience

**Undergraduate Researcher | Robotics Research Centre, IIIT Hyderabad** April 2025 – Present
*Research Assistant* *On-site*

- Designed and verified modular, low-latency C++ and Python software pipelines on Linux, implementing automated test harnesses and regression scripts to validate high-dimensional algorithms.
- Developed Python and Bash scripts to automate test generation, coverage tracking, and root-cause debugging across concurrent execution threads, ensuring deterministic and reproducible verification.

## Publications

**Towards Kinematic Actionable Infeasibility Detection in Motion Planning** | *IEEE RA-L (Accepted)* | arXiv

- Co-authored research introducing a formal verification and certification framework for proving path non-existence in high-DOF state spaces, leveraging parallel C++ and GPU-accelerated simplicial reconstruction.

## Projects

**5-Stage Pipelined RISC-V Processor** | *SystemVerilog, Verilog, Computer Architecture* | GitHub

- Designed microarchitecture and synthesizable RTL for sequential and 5-stage pipelined 32-bit RISC-V processors (IF, ID, EX, MEM, WB) with integrated data forwarding, hazard detection, and branch control logic.
- Developed self-checking SystemVerilog testbenches, assertion checkers, and directed/randomized instruction test suites to debug waveform failures and verify corner-case data, control, and structural hazards.

**4-Bit Carry Lookahead Adder (CLA) Design** | *Verilog HDL, CMOS, TSPC Logic, MAGIC, FPGA* | GitHub

- Implemented a high-speed 4-bit CLA across Verilog RTL, transistor-level static CMOS and TSPC dynamic logic, and custom physical layout in MAGIC using TSMC 180nm technology.
- Performed pre- and post-layout SPICE simulations, DRC/LVS checks, parasitic extraction, and FPGA hardware validation to verify functional correctness and optimize critical-path delay and power.

**ASIC RTL-to-GDSII Implementation & Verification** | *Verilog, Synthesis, STA, P&R, Python Scripting*

- Executed end-to-end RTL-to-GDSII physical design flow including logic synthesis, floorplanning, placement, Clock Tree Synthesis (CTS), routing, and DRC/LVS verification to achieve timing and power sign-off.
- Developed Python and Shell scripts to automate Static Timing Analysis (STA) report parsing and resolve setup/hold timing violations across multi-corner operating constraints.

**ML Model for VLSI Leakage Power Estimation** | *Python, Machine Learning, SPICE, Low-Power VLSI*

- Built a machine learning regression pipeline in Python to accurately estimate static subthreshold leakage power across digital CMOS circuits for energy-efficient hardware accelerator design.
- Modeled transistor-level leakage characteristics across four distinct operating zones (`NMOS_ON`, `NMOS_OFF`, `PMOS_ON`, `PMOS_OFF`), achieving high accuracy against golden SPICE simulations with significant speedup.

## Technical Skills

**Hardware & Verification:** SystemVerilog, Verilog HDL, UVM Concepts, RTL Design, Assertions (SVA), Functional Coverage, STA
**Languages & Scripting:** C, C++, Python (NumPy, SciPy, Pandas), Bash/Shell, TCL, Perl (Familiar), MATLAB, SQL
**VLSI & EDA Tools:** RTL-to-GDSII Flow, CMOS/TSPC Design, MAGIC, NGSpice, LTSpice, FPGA Prototyping, Synthesis
**Systems & Libraries:** Linux/Unix, Git, CMake, POSIX, Multi-threading, PyTorch, Scikit-Learn, Eigen

## Achievements/Extracurriculars

- Selected among the **Top 45 Teams** in the **Adobe University Hackathon 2026**.
- Secured **3rd Place** in **Megathon 2025**, building a Multi-Modal AI Solution for Estimating Vehicle Damage Claims.
- Achieved a peak rating of **1269** on **Codeforces**, solving 200+ algorithms and data structures problems.
- Secured **AIR 4391** in JEE Mains & **AIR 10931** in JEE Advanced; Cleared **KVPY 2022** & **Pre-RMO 2019**.

## Relevant Coursework

Digital VLSI Design | Intro to Processor Architecture | Digital Logic Design | Computer Architecture |
Data Structures and Algorithms | Operating Systems | Signal Processing | Probability & Random Processes | Linear Algebra